Who this document is for

This runbook covers the M2 Meshtastic nodes from first-time setup through replacing a broken unit. No deep technical background required.
Read each section fully before starting. If something goes wrong, every error has a fix listed.
M2 runs 2 rack-mounted Heltec V3 ROUTER nodes plus 8 Heltec T114 CLIENT loan devices.

The M2 mesh network provides off-grid radio communications at any event or deployment. Two infrastructure nodes in the rack relay messages between phones and the MQTT bridge. Eight T114 field nodes are loaned to attendees — each connects over LoRa to the rack nodes. Attendees join the channel by scanning a QR code. No cell service or internet required.

SECTION 1 WHAT YOU NEED

Item	Notes
8× Heltec T114 nodes	MF-1 through MF-8 — loan devices
8× USB-C data cables	Must be data cables — charge-only will not work
Windows or Linux PC with project folder	C:\SOURCE CONTROL\M2-Community-Node
T114 firmware .uf2 file	Download from meshtastic.org/downloads — Heltec T114
Python 3.12 + dependencies installed	pip install meshtastic pyserial qrcode[pil] reportlab
M2_SECRETS.md	For recording and rotating the PSK

Data cable vs. charge-only cable

If the config script says 'No port detected' after plugging in a node, the cable is probably charge-only. Try a different cable.
USB-C cables included with laptops and wall chargers are often charge-only.

SECTION 2 FLASH FIRMWARE — T114 FIELD NODES (DO ONCE PER NODE)

T114 uses a UF2 bootloader — no terminal or esptool needed. The node appears as a USB drive when in bootloader mode. Copy the firmware file onto the drive and it installs itself.

1 Download T114 firmware from meshtastic.org/downloads

- → `Select: Heltec T114`
- → `Download the .uf2 file`

2 Enter bootloader mode on the T114

- → `Double-tap the reset button quickly`
- → `A USB drive appears on your computer`
- → `If no drive appears: hold BOOT while plugging in USB`

3 Copy the firmware to the drive

→ Drag the .uf2 file onto the drive
→ Drive disappears automatically — node reboots with Meshtastic installed

4 Repeat for all 8 T114 nodes (MF-1 through MF-8)

How you know it worked
T114 E-ink screen shows the Meshtastic logo after reboot.
Each node is visible in the Meshtastic app over Bluetooth.

SECTION 3 CONFIGURE NODES

Flash vs. Configure — two completely different operations
FLASH (Section 2) — installs Meshtastic on the hardware. Done once per node, ever.
Never flash again unless doing a firmware upgrade.

CONFIGURE (this section) — sets the channel name, PSK, role, and name.
Run this before every event to rotate the PSK. No reflashing.

PSK must be rotated before every event
Anyone who scanned the QR at a past event still holds the old key.
Generate new key: python -c "import os,base64; print(base64.b64encode(os.urandom(16)).decode())"
Paste into CHANNEL_PSK_B64 in: config_m2_t114.py, config_heltec_v3.py, AND gen_m2_channel_qr.py
Save the new PSK to M2_SECRETS.md before running the script.

1 Run the T114 config script

```
python meshtastic/scripts/config_m2_t114.py
```

2 For each of the 8 T114 nodes — in slot order (MF-1 through MF-8):

→ Script prompts: '[1/8] Plug in node for: M2 Field Node 01'
→ Plug in the matching T114 via USB-C data cable
→ Press Enter — script pushes the PSK, channel, name, and role
→ Wait for '[OK] M2 Field Node 01 configured.'
→ Unplug and label. Repeat for MF-2 through MF-8.

How you know it worked
Terminal shows: [OK] M2 Field Node 01 configured.
Meshtastic app shows the node with channel name CommunityNode.

SECTION 4 GENERATE THE DAY-OF QR CARD

The QR card lets attendees join the CommunityNode channel by scanning with their phone. Generate a fresh card after every PSK rotation.

1 Run the QR generator

```
python meshtastic/scripts/gen_m2_channel_qr.py
```

2 Print the output file

→ Open: meshtastic/pdf/M2_Mesh_ChannelQR_DayOf.pdf
→ Print at 100% actual size — do NOT scale to fit
→ Cut on the dashed line — you get 2 identical cards per sheet
→ Place one at check-in. Give one to the lead organizer.

SECTION 5 PRINT ATTENDEE CONNECTION CARDS

Device checkout cards are front+back cards for each T114 loan node (MF-1 through MF-8). Print once, laminate, and reuse. Reprint only if node info or instructions change.

1 Generate the cards

```
python meshtastic/scripts/generate_m2_mesh_connection_card.py
```

2 Print double-sided

→ Open: meshtastic/pdf/M2_Mesh_DeviceCard_Front.pdf (front) + Back.pdf (back)
→ Print at 100% — cut on crop marks — 2 cards per sheet
→ Laminate each card for durability

SECTION 6 REPLACING A BROKEN NODE

If a T114 is lost or damaged, replace it with any spare T114 unit. Flash and configure it to match the slot of the broken node.

Step	What to do
1. Flash firmware	UF2 drag-and-drop (Section 2) — just the 1 replacement node
2. Configure it	python meshtastic/scripts/config_m2_t114.py — select its slot
3. Verify in app	Node should appear as MF-x in Meshtastic app over Bluetooth
4. Update device card	Label the replacement with its assigned slot number

SECTION 7 TROUBLESHOOTING

Problem	Fix
Bootloader drive not appearing	Hold BOOT while plugging in USB to force bootloader. Try a different cable.
T114 stuck in bootloader (drive stays mounted)	Firmware copy may have failed. Re-copy the .uf2 file onto the drive.
T114 GPS not acquiring fix	Move outdoors. GPS cold start takes 2-5 min. E-ink shows lock icon.
'No port detected' after plugging in T114	Use a data-capable USB-C cable. Try a different USB port.
pip install errors	pip install meshtastic pyserial qrcode[pil] Pillow reportlab
App can't find node in Bluetooth scan	Short-press reset. Toggle phone Bluetooth off and back on. Reopen app.
Nodes not messaging each other	Confirm all show CommunityNode channel and same PSK. Move nodes closer.
QR scan does nothing	Update Meshtastic app — older builds can't import URL channels.
Config script pushed wrong name to a node	Re-run config_m2_t114.py and redo only the affected slot.
T114 E-ink screen stays blank after reboot	E-ink refresh takes 10-15 seconds. Wait, then verify in app.

SECTION 8 QUICK REFERENCE — FULL WORKFLOW

Command	Purpose
python meshtastic/scripts/config_m2_t114.py	Push PSK, channel, name, and role to all 8 T114 field nodes
python meshtastic/scripts/gen_m2_channel_qr.py	Generate the day-of QR card for attendees
python meshtastic/scripts/generate_m2_mesh_connection_car	Generate T114 device checkout cards (8 nodes, front+back)
python meshtastic/scripts/generate_m2_mesh_handoff.py	Rebuild the operator handoff reference PDF
python meshtastic/scripts/generate_m2_mesh_runbook.py	Rebuild this runbook PDF
python meshtastic/scripts/generate_m2_mesh_cheatsheet.py	Rebuild the 1-page operator cheatsheet

Normal pre-event checklist (nodes already configured from a prior event)

1. Generate new PSK — save to M2_SECRETS.md
2. Update CHANNEL_PSK_B64 in config_m2_t114.py, config_heltec_v3.py, AND gen_m2_channel_qr.py
3. Push new config to T114 nodes — python meshtastic/scripts/config_m2_t114.py
4. Regenerate QR card — python meshtastic/scripts/gen_m2_channel_qr.py
5. Print QR cards — meshtastic/pdf/M2_Mesh_ChannelQR_DayOf.pdf
6. Test 2 phones: Bluetooth pair → scan QR → send message
7. Verify MQTT bridge running on Pi 2 before deployment